Python data science

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Recap

- Great job!
- We only have 4 days to go from introductory to advanced Python concepts.
  - Plus: programming tools like VSCode and `git`!
- Classes and functions can be abstract, but they are the building blocks of what we will do today.
  - Hopefully today is more familiar to people who have used R!

Today: Python data science

- Introduction to Python data science tools.
- Introduction to a basic data science workflow.
- This afternoon: collaborating on a data science project.

Tomorrow

- Data science “challenge”
- *Predicting the nightly price of AirBnBs in London*
- See the Guidelines here.

Data science

- Definition of data science:
  - *“Extracting meaningful insights from data.”*
- *Meaningful* is important.
  - Use the tools of programming / statistics to create meaning from your data.
- Usually, there is no “right” answer, just “better” and “worse” answers.
  - You exercise a lot of judgement.

Data science workflow

- Data science is **not** just machine learning.
  - Most data science work is:
    - Data preparation
    - Data transformation
    - Method selection
      - *Statistics / machine learning*
    - Communicating results

Python data science tools

- Today, we will learn about the most popular Python data science “stack”
  - Data preparation / data transformation
    - `pandas`, `numpy`
  - Statistics / machine learning
    - `sklearn`
  - Communicating results
    - `matplotlib`

Python data science tools

- Tomorrow, we will use this “stack” to do our data science project
- Exploratory analysis, data transformation
  - `pandas`
- Regression model fitting and evaluation
  - `sklearn`
- Visualize results
  - `pandas`, `matplotlib`

R equivalents

- Python libraries mostly have their R equivalents:
  - `pandas` : `dplyr`
  - `matplotlib` : `ggplot2`
  - `sklearn` : `caret`?
- See what you prefer, I use both languages!

Diving deeper

- Python has many other options for data science tools.
- Alternatives to `pandas`:
  - `polars` *(Like Python's version of* `data.table`*)*
  - `dask`
- Alternatives to `sklearn`:
  - …?
- Alternatives to `matplotlib`:
  - `seaborn`
  - `plotnine` *(R users might like this one!)*

Tutorial #1: Selecting data

- pandas-cookbook: Selecting data (Chapter 2)
- Core concepts:
  - Reading data from a `.csv` file
  - Inspecting a dataset
  - Selecting data

Tutorial #2: More selecting data

- pandas-cookbook: More selecting data (Chapter 3)
- Core concepts:
  - Selection by multiple columns
  - The role of `numpy` in `pandas`
  - Basic plotting (`matplotlib` in `pandas`)

Data: Tutorials 1 and 2

- Tutorials #1 and #2 come from the pandas-cookbook.
- Go to the `/data` folder in the GitHub repository (link above).
- Download the `311-service-requests.csv` file and store it on your computer.

Packages: Tutorials 1 and 2

- Install the required packages using your terminal in VSCode

```
pip install pandas matplotlib scikit-learn
```

- Trouble installing? Tell me!